End-to-End Supply Chain Analysis

Business Performance, Inventory & Shipment Report

KPI Scorecard

Metric	Value	Status
Total Orders	30,871	Informational
Total Revenue	$6,181,476.00	Informational
Total Profit	$3,994,192.00	Informational
Profit Margin	64.62%	Healthy
Avg Shipment Delay	3.07 days	Needs Attention
Orders Delayed 4 Days	58.3%	Critical
Total Storage Cost	$86,430.92	Informational
Inventory Surplus	4,647 units	Needs Attention

Key Findings

1. 100% of orders experience shipment delays ranging from 1 to 4 days, with 58.3% experiencing the maximum 4-day delay. This indicates a systemic logistics issue.

2. Fan Shop is the highest-performing department generating $1.64M in profit, followed by Apparel ($912K) and Golf ($655K).

3. Apparel and Golf departments show the largest inventory surplus, indicating significant overstocking that increases carrying costs.

4. The Perfect Fitness Perfect Rip Deck is the single most profitable product at $630,924 in total profit.

5. Revenue and profit remained stable from 2015 to mid-2017, followed by a sharp decline in late 2017 requiring further investigation.

Analysis Charts

Chart 1: Shipment Delay Distribution

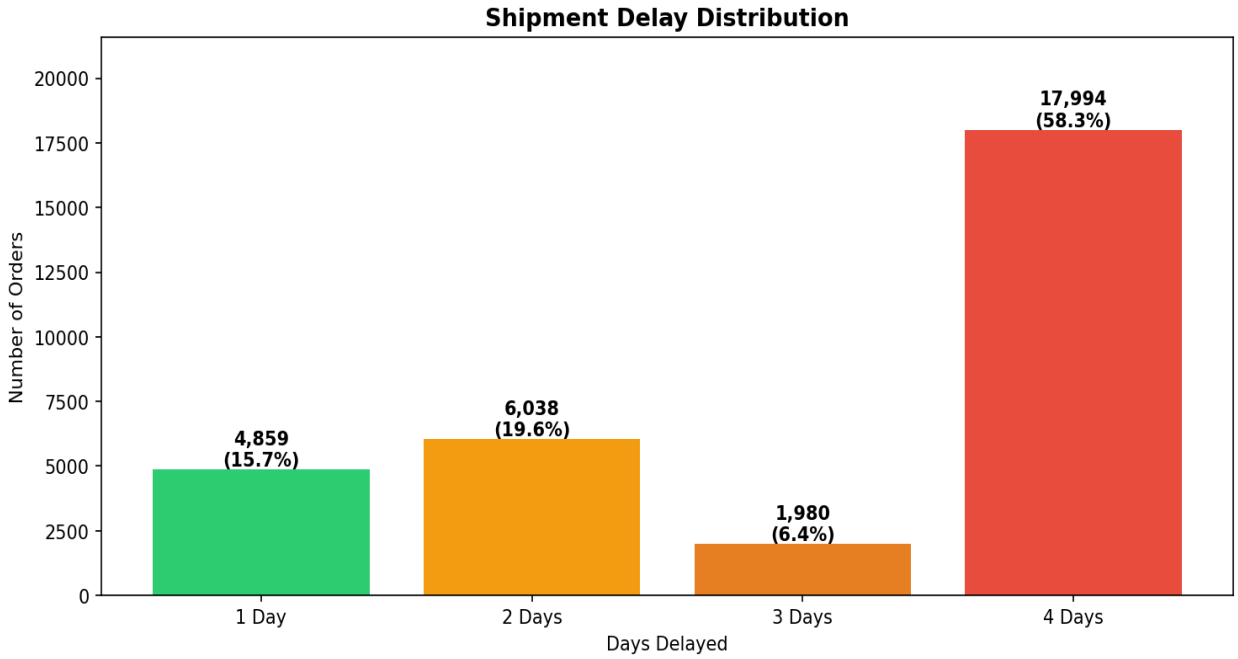


Chart 2: Profit by Product Department

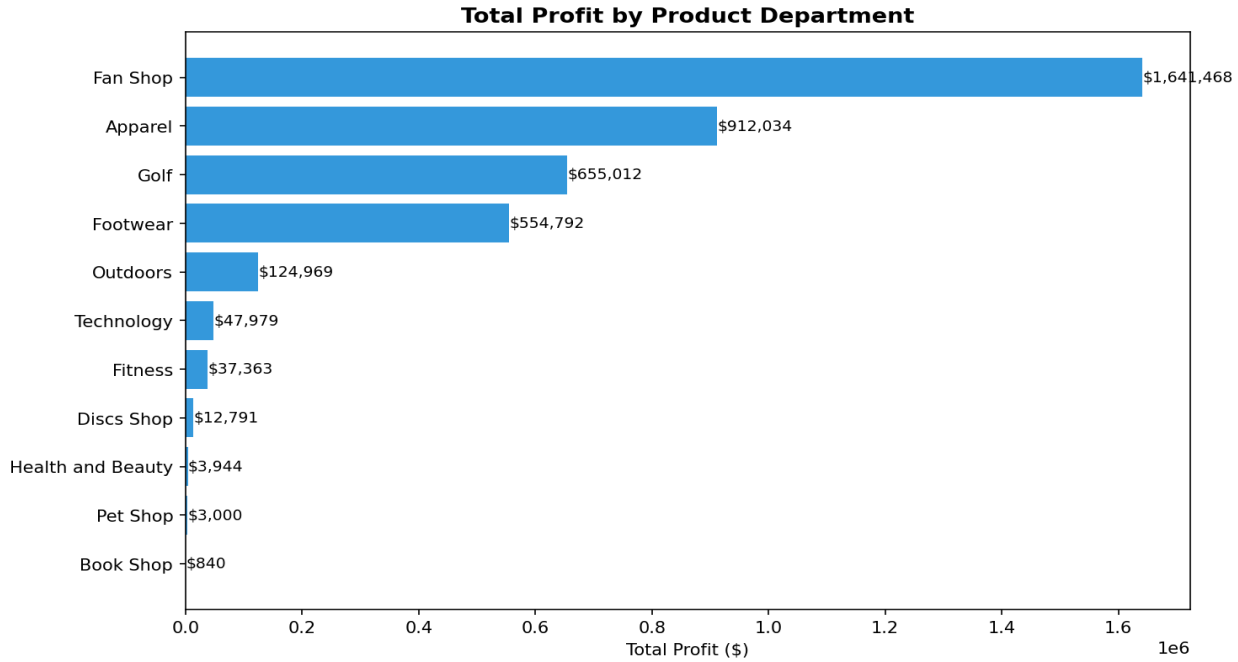


Chart 3: Inventory Gap by Department

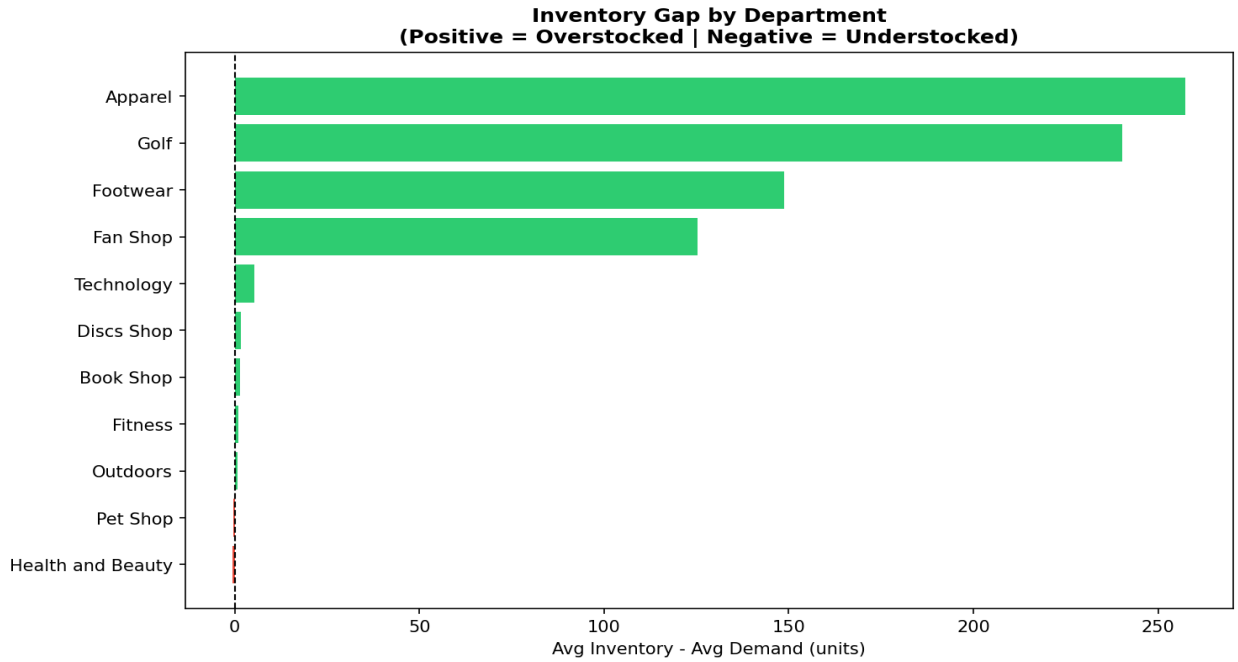
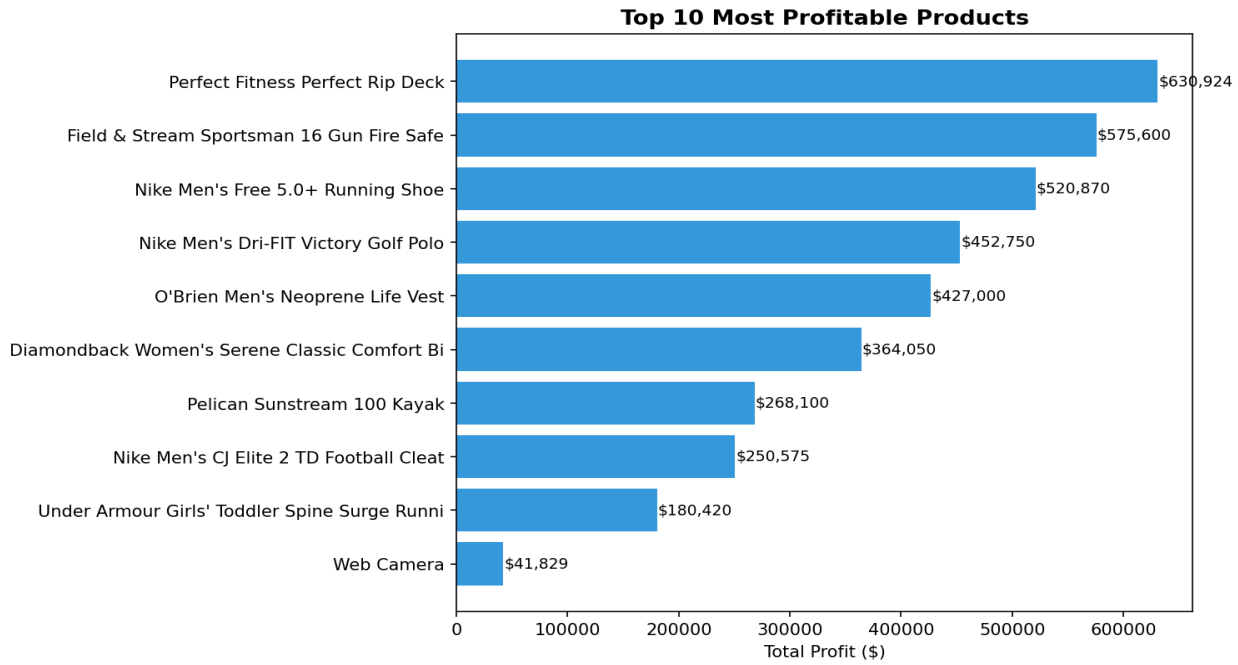


Chart 4: Top 10 Most Profitable Products



Recommendations

1. Investigate Shipment Delays: All orders are delayed by 1-4 days. Conduct a root cause analysis on logistics partners, warehouse processing times, and shipping mode efficiency.

2. Optimize Inventory Levels: Apparel and Golf departments are significantly overstocked. Reduce purchase orders and run targeted promotions to clear surplus inventory.

3. Focus on High-Margin Products: Prioritize stock availability for top performers like Perfect Fitness Rip Deck, Field & Stream Gun Safe, and Nike Running Shoes.

4. Investigate 2017 Revenue Decline: The sharp drop in late 2017 requires further analysis to determine whether it is seasonal, market-driven, or operational.

5. Review Low-Profit Departments: Book Shop, Pet Shop, and Health & Beauty generate minimal profit. Evaluate whether to expand, reposition, or discontinue these product lines.